



[EC2102]: Digital Design and Computer organization

Teaching Scheme: TH: -2 Hours/Week PR: -2 Hours/Week	Credit TH:3	Examination Scheme: In Sem. Evaluation:20 Marks Mid Sem. Exam :30 Marks End Sem. Exam :50 Marks
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Course Prerequisites: Logic gates and Boolean algebra, logical equation reduction technique, K-MAP.

Course Objective:

The course is served to acquaint the students with the fundamental principles of digital logic and various devices used to implement logical operations. The course provides foundation for further studies in areas such as Communication, VLSI, Computer, Microprocessor. To acquaint the students with the fundamental principles of computer and various devices used to implement logical operations.

To acquaint the students with the functions of input/output devices and memory

Course Outcome:

After successful completion of the course, students will able to:

CO1: Outline features of the basic logic circuit, modules of computer

CO2: Summarize functionality of the digital logic circuit and various modules of computer.

CO3: Design combinational and sequential circuits for given task.

Course Contents

UNIT-I	Combinational Logic Design	07Hours
Standard representations for logic functions, minimization of logical functions using k map (up		

to 4 variables), don't care conditions. MSI devices like Comparators, Multiplexers, Encoder , Decoder, Driver & Multiplexed Display, Half and Full Adders, Subtractors, Serial and Parallel Adders, BCD Adder, Barrel shifter and ALU , Decoder, Driver & Multiplexed Display.

UNIT-II	Sequential Logic Design	07Hours
1 Bit Memory Cell, Clocked SR, JK, MS J-K flip flop, D and T flip-flops. Use of preset and clear terminals, Excitation Table for flip flops. Conversion of flip flops. Application of Flip flops: Registers, Shift registers, Counters (ring counters, twisted ring counters), Sequence Generators, ripple counters, up/down counters, synchronous counters, lock out, Clock Skew, Clock jitter. Effect on synchronous designs.		
UNIT-III	State Machines	07 Hours
Basic design steps- State diagram, State table, State reduction, State assignment, Mealy and Moore machines representation, Implementation, finite state machine implementation, Sequence detector. Introduction to Algorithmic state machines - Construction of ASM chart and realization for sequential circuits.		
UNIT-IV	Basic structure of computer	07 Hours
Computer types, functional units- input unit; output unit; ALU; control unit; memory unit, Basic operational concepts, Bus structure, Performance-processor clock; basic performance equation; pipelining & superscalar; operation; clock rate; instruction set: CISC & RISC; Multiprocessors & Multi computers		
UNIT-V	Arithmetic & control unit	06 Hours
Addition & subtraction of signed binary number, signed operand multiplication, Booths algorithm, Integer division, single bus organization- register transfer; performing an arithmetic or logic operation; fetching and storing word from/to memory; execution of complete instruction; branch instruction, multi bus organization.		
UNIT-VI	Memory & input-output organization	07 Hours
I/O organization – accessing I/O devices, Interrupts- interrupt hardware, enabling and disabling interrupts, handling multiple requests, Direct memory access- bus arbitration, Buses- Synchronous; asynchronous, semiconductor RAM memories-internal organization of memory chips; static memories; asynchronous and synchronous DRAM; structure of large memories, cache memory, virtual memory.		
Text Books: 1. R.P. Jain , “Modern digital electronics” , 3rd edition , 12th reprint Tata McGraw Hill Publication, 2007. 2. M. Morris Mano, “Digital Logic and Computer Design” 4th edition, Prentice Hall of India, 2013.		

3. Zvonco Vranesic and Safwat Zaky, "Computer Organization" 5th edition Tata McGrail
4. John P Hayes, "Computer Architecture and Organization" 3rd edition Prentice Hall of India

Reference Book:

1. Anand Kumar, "Fundamentals of digital circuits" 1st edition, Prentice Hall of India, 2001
2. William Stallings, "Computer Organization and Architecture: Designing for Performance" 2nd edition Prentice Hall of India

PRSCOE

**JSPM's
RAJARSHI SHAHU COLLEGE OF ENGINEERING
TATHAWADE, PUNE-33**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

NPTEL Video Links

Subject: Digital System Design

Unit 1:

<https://www.youtube.com/watch?v=CeD2L6KbtVM&list=PL803563859BF7ED8C>

<https://www.youtube.com/watch?v=sUutDs7FFeA&list=PL803563859BF7ED8C&index=3>

<https://www.youtube.com/watch?v=XCiLHOZsQl8&list=PL803563859BF7ED8C&index=4>

<https://www.youtube.com/watch?v=ygm25sqgepg&list=PL803563859BF7ED8C&index=7>

<https://www.youtube.com/watch?v=ygm25sqgepg&list=PL803563859BF7ED8C&index=7>

<https://www.youtube.com/watch?v=kgL5UaSVuro&list=PL803563859BF7ED8C&index=9>

Unit 2:

<https://youtu.be/5-PI4T25OXI>

Unit 3:

https://youtu.be/2ecMG_OciLo

<https://youtu.be/4CRPlaBnfV0>

Course Coordinator